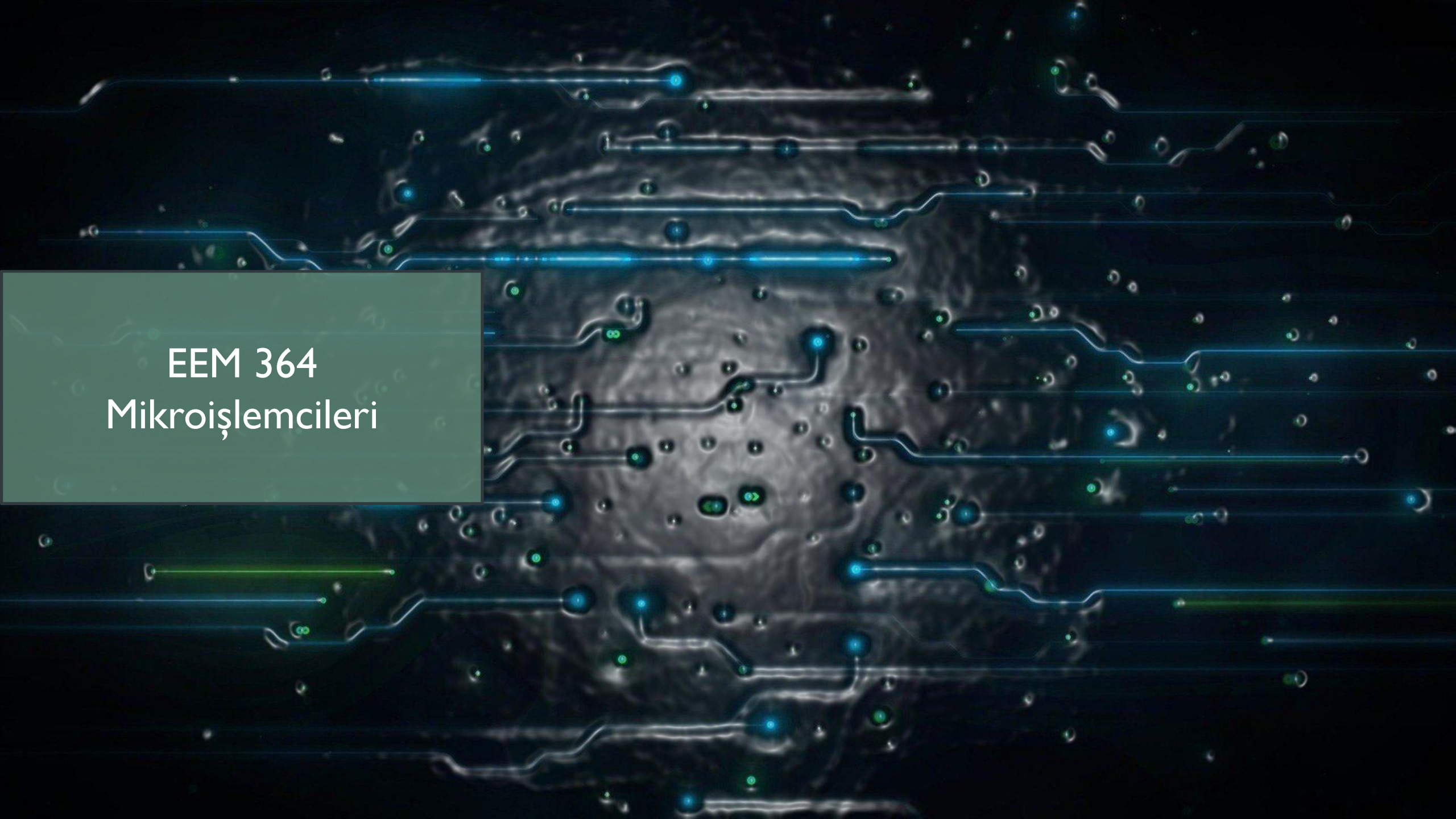

EEM 364
Mikroişlemcileri

Hafta 11
Prosedürler

```
.MODEL SMALL

.DATA

x db 1,2,3,4,5,6,7,8,9,10

tc db ?
tt db ?

.CODE

mov ax , @data
mov ds , ax



mov si , 0
mov di , 1
mov cx , 5
mov ax , 0

TPL: add al , x[si]
     add ah , x[di]
     add si , 2
     add di , 2
     dec cx
     jnz TPL

mov tc , al
mov tt , ah


hlt
```

```
.MODEL SMALL

.DATA

T db 8
F db 0

.CODE

mov ax , @data
mov ds , ax


mov ax , 0

mov al , 0
mov ah , 1
mov bl , 0

mov cl , T

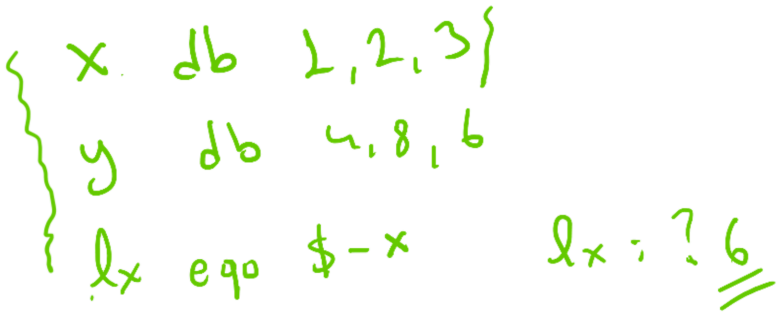

FIB: mov bl , ah
     add ah , al
     mov al , bl

     mov F ,  ah
     dec cl
     jnz FIB




hlt
```

Assembly Dili Yapısı: Etiketler ve Döngüler



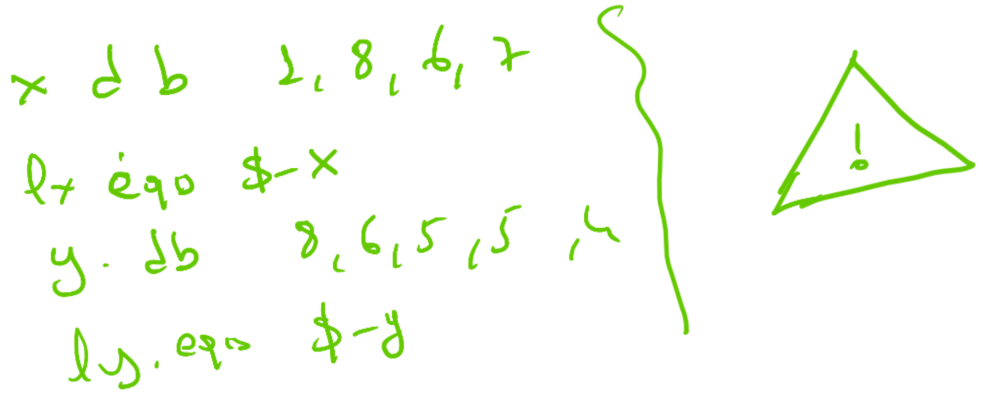



```
01 .MODEL SMALL
02
03 .DATA
04
05
06 x  db 1, 2, 15, 33, 66, 5, 3, 8, 9, 25, 34, 22
07 lx equ $-x
08 ts db 25
09
10
11 U db ?
12 V db ?
13
14 .CODE
15
16 mov ax , @data
17 mov ds , ax
18
```

100....

$: bir önceki dizinin son adresi

- Basit segment tanımlama ve
- EQU direktifi



Koşullar ve Döngüler

CMP → SUB AX, BX

- JNZ/JZ
- INC
- DEC
- OFFSET
- EQU
- LEA
- CMP
- JE , JG , JL

```
CMP register, register;
CMP AX, BX
CMP register, memory;
CMP AX, SONUC
CMP register,
    immediate;
CMP AX,5
```

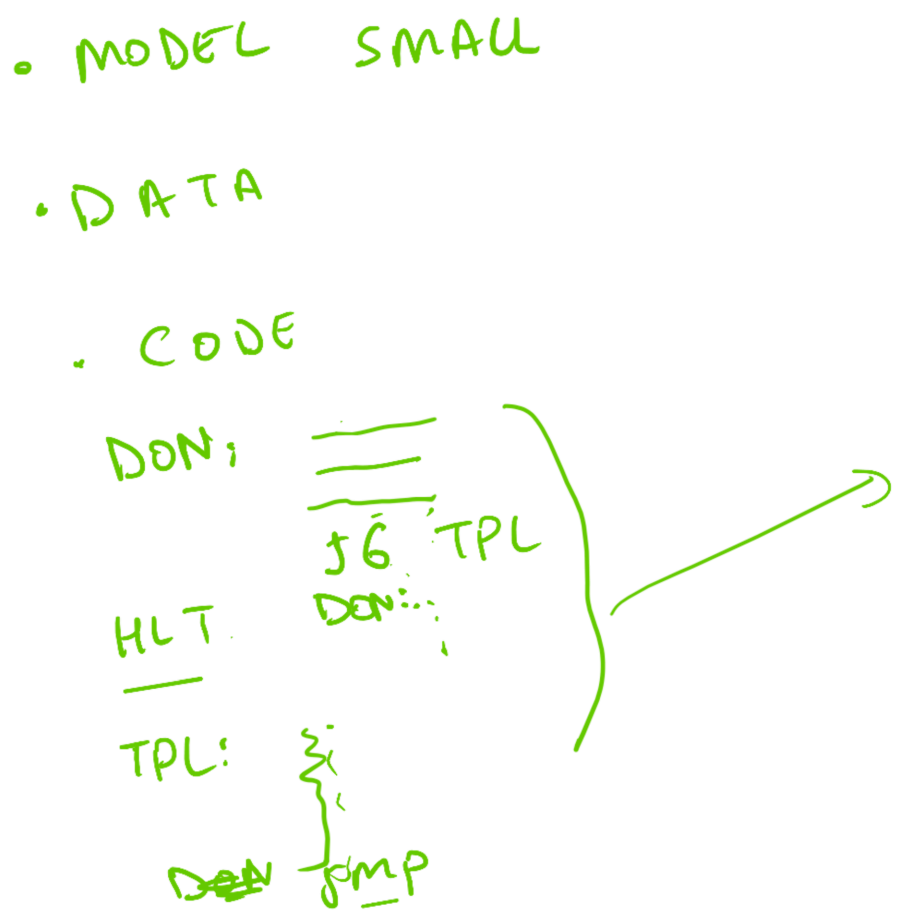
```
CMP memory, register;
CMP SONUC, AX
CMP memory, immediate;
CMP SONUC,5
```

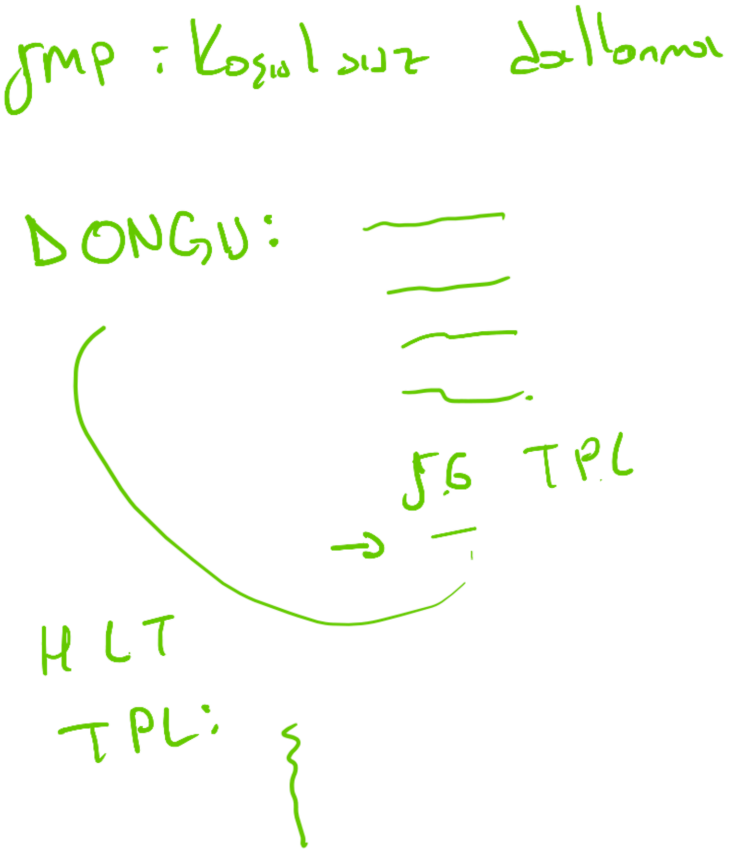
CMP sadece bayrak değiştirir

https://web.karabuk.edu.tr/emelkocak/indir/MTM305/Kontrol_Komutlar%C4%B1.pdf

Koşullar ve Döngüler

Koşul İfadeleri Yazma Yöntemi





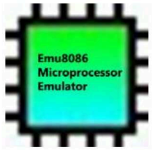
Koşullar ve Döngüler

- JNZ/JZ
- CMP
- INC
- DEC
- OFFSET
- EQU
- LEA
- CMP
- JE , JG , JL
- TEST

Bit Tabanlı AND İşlemi

```
01  .MODEL SMALL
02
03  .DATA
04
05  .CODE
06
07  mov ax , @data
08  mov ds , ax
09
10
11
12  mov al , 5
13  mov bl , 8
14
15  test al , 01H
16  test bl , 01H
17
18
19  hlt
```

Sayıları çift ve tek olma durumuna göre toplama



```
.MODEL SMALL

.Data
    dizi DB 1,3,0,2,2,3,1,0,1,2,0,3,1,0,2

    sifir_sayaci DB 0
    bir_sayaci   DB 0
    iki_sayaci   DB 0
    uc_sayaci    DB 0
.Code

    MOV AX, @Data
    MOV DS, AX

    MOV SI, 0
    MOV CX, 15
    XOR AX, AX
    XOR BL, BL

    MOV BP, 0

DONGU:
    MOV AL, dizi[SI]

    CMP AL, 0
    JE SAY_0

    CMP AL, 1
    JE SAY_1

    CMP AL, 2
    JE SAY_2

    CMP AL, 3
    JE SAY_3

D:

    inc si
    dec cx
    jnz DONGU
```

```
HLT

SAY_0:
    INC sifir_sayaci
    JMP D

SAY_1:
    INC bir_sayaci
    JMP D

SAY_2:
    INC iki_sayaci
    JMP D

SAY_3:
    INC uc_sayaci
    JMP D
```

1011

1011

0060

```
.MODEL SMALL

DATA_SEGMENT SEGMENT

dizi DB -5, 4, -7, 2, -3, 1, 6, -2, 0, -4

DATA_SEGMENT ENDS

CODE_SEGMENT SEGMENT

ASSUME CS:CODE_SEGMENT, DS:DATA_SEGMENT

BASLA:

    MOV AX, DATA_SEGMENT
    MOV DS, AX


    MOV SI ,  0
    MOV CX , 10
    XOR AX , AX

    XOR BL , BL

TOPLA:
    MOV AL, dizi[SI]


    CMP AL, 0
    JS NEG_TOP

    ADD BL , AL

DON:
    INC SI
    DEC CX
    JNZ TOPLA

     |
HLT

NEG_TOP:
    NEG AL
    ADD BL, AL


    JMP DON

CODE_SEGMENT ENDS
```



LAB ÖDEVİ

- x db 1,2,4,5,7,1 y db 3,1,5,2,2,4

$$\sum |x(i) - y(i)|$$

- Palindrom sayı ve yazı tespiti yapan kod
- 12321 , KABAK vb.

```
01 .MODEL SMALL
02
03 .DATA
04
05 T db 8,34,5,67,8,86,234
06
07 F db  ($-T)
08 F equ ($-T)
09
10 .CODE
11
12 mov ax , @data
13 mov ds , ax
14
15
16
17 hlt
```

Sayılar ile 4 işlem olabilir

LAB ÇÖZÜMLERİ

```
.MODEL SMALL

.DATA
    x DB 'KABddAK'
    uzunluk DB 5
    sonuc   DB 0

.CODE

START:
    MOV AX, @DATA
    MOV DS, AX

    MOV SI, 0
    MOV CL, uzunluk
    DEC CL
    MOV DI, CX

    MOV BL, 1

KARSILASTIR:
    MOV AL, x[SI]
    MOV AH, x[DI]
    CMP AL, AH
    JNE DEGIL

    INC SI
    DEC DI
    CMP SI, DI
    JL KARSILASTIR

    JMP SONUC_YAZ

DEGIL:
    MOV BL, 0

SONUC_YAZ:
    MOV sonuc, BL
    HLT

END START
```

```
.MODEL SMALL

.DATA
    x DB 1,2,4,5,7,1          ; x dizisi
    y DB 3,1,5,2,2,4          ; y dizisi
    N DB 6                    ; Eleman sayisi
    T DB 0                    ; Toplam sonuc burada saklanacak

.CODE

START:
    MOV AX, @DATA
    MOV DS, AX

    XOR BX, BX                ; BX = toplam (baslangicta sifirla)
    MOV SI, 0                 ; Dizi indexi

    MOV CL, N                 ; Dongu sayaci

LOOP_START:

    MOV AL, x[SI]             ; AL = x[i]
    MOV AH, y[SI]             ; AH = y[i]

    SUB AL, AH                ; AL = x[i] - y[i]
    JS  NEG_TOP
    JNS POS_TOP

DEVAM:

    INC SI
    DEC CL
    JNZ LOOP_START

    HLT

POS_TOP:
    ADD T , AL
    JMP DEVAM

NEG_TOP:
    NEG AL
    ADD T , AL
    JMP DEVAM
```

Prosedürler

- JNZ/JZ
- CMP
- INC
- DEC
- OFFSET
- EQU
- LEA
- CMP
- JE , JG , JL
- TEST
- PROC

```
.MODEL SMALL

.DATA
.CODE


main proc far

      mov ax , @data
      mov ds , ax

      call F1

      mov ax , bx


      hlt
main endp


F1 PROC

     mov bx , 1233H
     ret
F1 ENDP
```

Prosedürler

- JNZ/JZ
- CMP
- INC
- DEC
- OFFSET
- EQU
- LEA
- CMP
- JE , JG , JL
- TEST
- PROC
- LOOP

```
 .MODEL SMALL

.DATA
    x  db 1, 2, 15, 33, 66, 5, 3, 8, 9, 25, 34, 22
    lx equ $-x
    T db ?

.CODE


main proc far

     mov ax , @data
     mov ds , ax

     call F1

     mov T , al

     hlt

main endp


F1 proc

    mov cx , lx
    mov al , 0

    lea si , x

    L:
      add al , [si]
      inc si
      loop L

    ret

F1 endp
```

Prosedürler

- JNZ/JZ
- CMP
- INC
- DEC
- OFFSET
- EQU
- LEA
- CMP
- JE , JG , JL
- TEST
- PROC
- LOOP

```
 .MODEL SMALL

.DATA
    x  db 1, 2, 15, 33, 66, 5, 3, 8, 9, 25, 34, 22
    lx equ $-x
    T db ?

.CODE


main proc far

     mov ax , @data
     mov ds , ax

     call F1

     mov T , al

     hlt

main endp


F1 proc

    mov cx , lx
    mov al , 0

    lea si , x

    L:
      add al , [si]
      inc si
      loop L

    ret

F1 endp
```

Prosedürler

Bir dizinin min veya max elemanlarını bulan prosedürü yazınız

- JNZ/JZ
- CMP
- INC
- DEC
- OFFSET
- EQU
- LEA
- CMP
- JE , JG , JL
- TEST
- PROC
- LOOP

Prosedürler

Çarpma ve Bölme İşlemi

- JNZ/JZ
- CMP
- INC
- DEC
- OFFSET
- EQU
- LEA
- CMP
- JE , JG , JL
- TEST
- PROC
- LOOP

```
.MODEL SMALL

.DATA

A equ  15
C equ  4

Z dw 0

.CODE

main proc far

    mov ax , @data
    mov ds , ax


    mov ax , 0
    mov bx , 0

    call carp

    mov Z , bx

    HLT

main endp

carp proc

    mov al , A
    mov cx , C


    CAR: add bx , ax
         loop CAR


    ret

carp endp
```

```
.MODEL SMALL

.DATA

X equ  53
B equ  7


Z dw 0


.CODE

main proc far

    mov ax , @data
    mov ds , ax

    mov ax , 0

    call bol

    hlt

main endp


bol proc

    mov al , X
    mov bx , 0

    L:   inc ah
         sub al , B
         cmp al , B
         jge L
    ret

bol endp
```

Prosedürler

Sıralama Algoritması

- JNZ/JZ
- CMP
- INC
- DEC
- OFFSET
- EQU
- LEA
- CMP
- JE , JG , JL
- TEST
- PROC
- LOOP

```
.MODEL SMALL

.DATA

x   db 1,3,5,2,2,1,8,0,4, 7
lx equ $-x-1

.CODE

main proc far
    mov ax , @data
    mov ds , ax

L0:
    mov ax , 0
    mov cx , lx
    mov si , 0
    mov dl , 0


L:
    mov al , x[si]
    mov ah , x[si+1]

    call SWP
    inc si
    loop L

    cmp dl , 0

    jnz L0

HLT

main endp

SWP proc

    cmp al , ah
    jle not_swp
    mov x[si+1] , al
    mov x[si]   , ah

    inc dl

    not_swp:
    ret
SWP endp
```

Prosedürler

ÇARPMA KOMUTU

- **JNZ/JZ**
- **CMP**
- **INC**
- **DEC**
- **OFFSET**
- **EQU**
- **LEA**

- **CMP**
- **JE , JG , JL**
- **TEST**
- **PROC**
- **LOOP**
- **MUL**

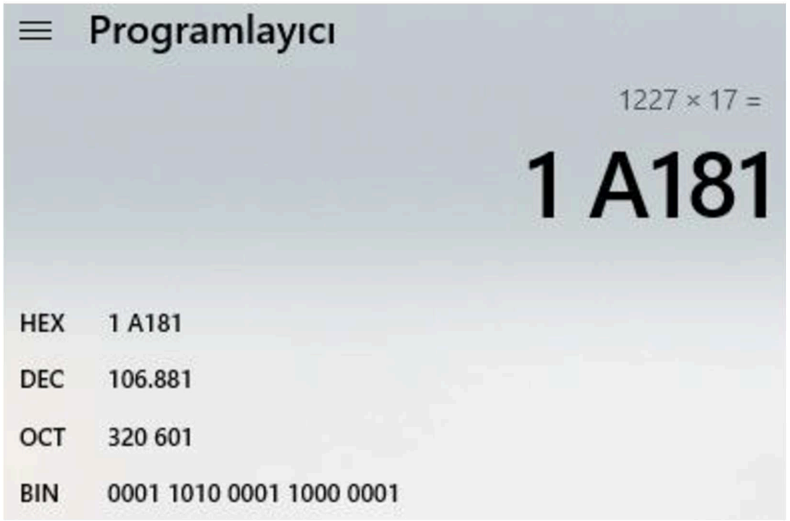
- **MUL ve IMUL**
- reg , reg
- reg , mem

- **Çarpılan**:
  - AL veya AX
- **Çarpan**:
  - BL,CL,DL
  - BX,CX,DX

Prosedürler

ÇARPMA KOMUTU

```
MOV AX , 1227H
MOV DX , 17H
MUL DX
```



- **MUL ve IMUL**
- reg , reg
- reg , mem

- **Çarpılan**:
  - AL veya AX
- **Çarpan**:
  - BL,CL,DL
  - BX,CX,DX

YAZMAÇLAR	
AH: 12	AL : 27
AX :A1 81	
DX: 00 01	

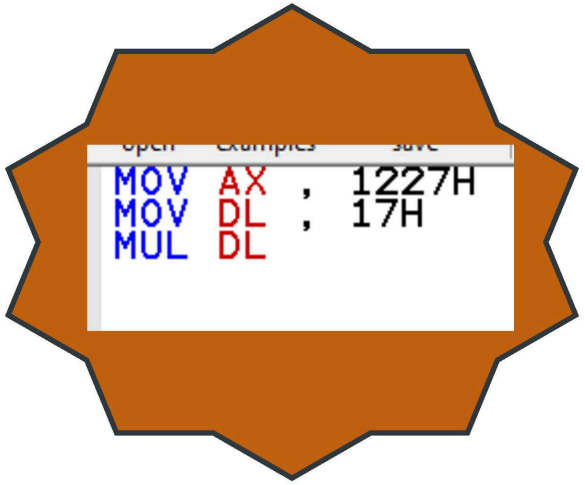
Prosedürler

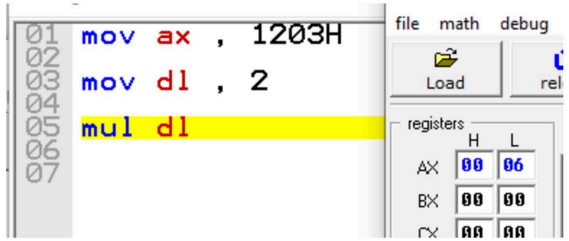
ÇARPMA KOMUTU

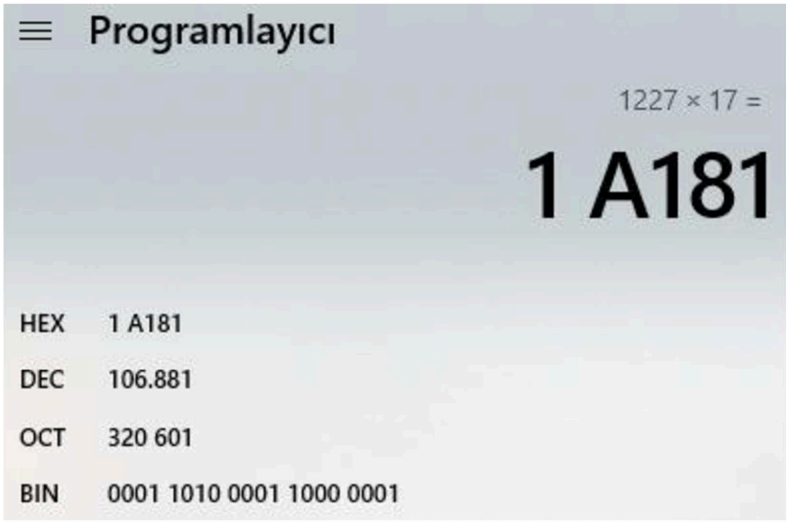
- **MUL ve IMUL**
- reg , reg
- reg , mem

- **Çarpılan**:
  - AL veya AX
- **Çarpan**:
  - BL,CL,DL
  - BX,CX,DX

```
MOV AX , 1227H
MOV DX , 17H
MUL DX
```







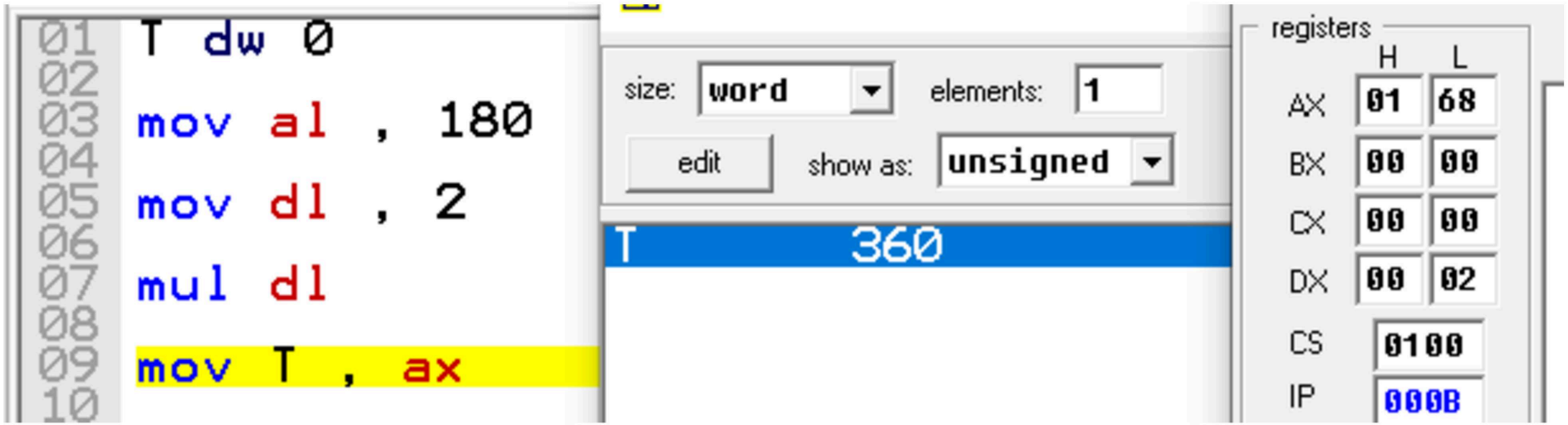
YAZMAÇLAR	
AH: 12	AL : 27
AX :A1 81	
DX: 00 01	

Prosedürler

ÇARPMA KOMUTU

- **MUL ve IMUL**
- reg , reg
- reg , mem

- **Çarpılan**:
  - AL veya AX
- **Çarpan**:
  - BL,CL,DL
  - BX,CX,DX



Prosedürler

ÇARPMA KOMUTU

- **MUL ve IMUL**
- reg , reg
- reg , mem

- **Çarpılan**:
  - AL veya AX
- **Çarpan**:
  - BL,CL,DL
  - BX,CX,DX

```
01 .MODEL SMALL
02
03 .DATA
04
05 X equ  6
06
07 S dw 0
08
09
10 .CODE
11
12 mov ax , @data
13 mov ds , ax
14
15
16 mov ax , 0001H
17 mov cx , X
18
19 L:
20 mul cx
21 loop L
22
23 mov S , ax
24
25
26 HLT
27
```

Faktöriyel !

Prosedürler

BÖLME KOMUTU

- **DIV ve IDIV**
- reg , reg
- reg , mem

```
MOV AL , 27H ; 39
MOV DL , 17H ; 23
DIV DL
```

YAZMAÇLAR	
AH: ~	AL : 27
AX: 10 01	

- **Bölünen**:
  - AL veya AX
- **Bölen**:
  - BL, CL, DL
  - BX, CX, DX

```
MOV AL , 42H; 66
MOV DL , 12H; 18
DIV DL
```

YAZMAÇLAR	
AH: ~	AL : 42
AX: 0C 03	

Prosedürler

BÖLME KOMUTU

- **DIV ve IDIV**
- reg , reg
- reg , mem

- **Bölünen**:
  - AL veya AX
- **Bölen**:
  - BL, CL, DL
  - BX, CX, DX

```
.MODEL SMALL

.DATA
x equ 53
y equ x/2

BN db  0
BM db 0
FL db 1

.CODE

main proc far

mov ax , @Data
mov ds , ax

mov cl , 2
mov dx , x

call asal


hlt
main endp

asal proc

    L:
    mov ax , dx
    div cl
    cmp ah , 0
    jz not_asal
    inc cl
    cmp cl , y
    jnz L


    D:

    ret

    not_asal:

    mov BN , cl
    mov BM , al
    mov FL , 0

    jmp D

asal endp
```

Asal Sayı Bulma